

Online Retail Dataset: Data Cleaning, Exploratory Analysis & Sales Insights

Objective

The purpose of this project was to clean and analyze the Online Retail dataset using SQL Server and Power BI to identify sales patterns, customer purchasing behavior, and revenue trends while improving overall data quality.

Dataset Overview

Metric, Value

Total Rows, 541,910

Total Columns, 8

Missing Values, 136,534

Primary Key, Invoice No

Main Focus, Retail Transactions

Data Cleaning Process

The following cleaning operations were performed:

- Imported dataset from Excel into SQL Server in CSV format

- Corrected data types for quantity, prices, and invoice dates
- Converted invoice dates into DATETIME format
- Standardized inconsistent text fields
- Replaced missing customer IDs and descriptions
- Removed duplicate transaction records
- Identified negative quantities and prices
- Validated transaction integrity
- Exported dataset into Excel
- Imported dataset into Power BI to perform DAX operations.

Exploratory Data Analysis (EDA)

The dataset was explored to analyze:

- Revenue trends over time
- Revenue contribution by country
- Top 5 Customers with Product quantity distribution

Key Performance Indicators

- Total Revenue - \$8.2m
- Total Orders – 22k
- Returned Orders – 8,680
- Total Quantity Sold - 5m

Visualizations Included

- Line Chart — Monthly Revenue Trend
- Bar Chart — Revenue by Country
- Column Chart — Most Customers with Product Quantity
- Pie Chart – Country by Order Status
- Donut Chart – Percentage of Orders by Order Status.
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Key Insights

- The United Kingdom generated the highest revenue among all countries
- A small group of products contributed significantly to total sales.
- Negative quantities represented returned or cancelled transactions.
- Sales increased during holiday periods and peak shopping seasons.
- Most products had low unit prices but were purchased in high quantities.

Business Implications

- Revenue is heavily concentrated in specific countries and products.
- Seasonal trends can support inventory planning and forecasting.
- Product return patterns may indicate customer satisfaction or operational issues.

Conclusion

The Online Retail dataset was successfully cleaned and transformed for analytical purposes. The analysis uncovered important sales and customer behavior patterns that can support retail decision-making, revenue optimization, and operational planning.

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